

USA Real Estate Market Analysis
Capstone 3 Final Report
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Problem Identification:

The US residential real estate market contains millions of active listings but pricing varies widely by state, city and ZIP code. Buyers trying to decide whether a listing is fairly priced, agents setting list prices for sellers and investors screening markets all face the same core question. Given what I know about a property and where it is located, what should it cost? The goal of this project is to build a model that predicts residential listing price from property features and location and to identify which markets offer the best value.

Data:

I used four data sources. The main dataset is the USA Real Estate Dataset from Kaggle, which contains 2.2 million residential listings scraped from Realtor.com covering all 50 states. Each listing includes price, bedrooms, bathrooms, house size, lot size, city, state, ZIP code and sale status. I added economic context from two Census tables. ACS DP03 provided median household income, unemployment rate and poverty rate by ZIP code. ACS S2501 provided occupied housing units by ZIP code. Both use 2019-2023 5 year estimates. I also used Census State to State Migration Flows for 2019, 2021, 2022, 2023 and 2024 to track which states are gaining and losing people.

Data Wrangling:

I cleaned the listings by dropping rows missing price, state, or ZIP code because those columns are essential. I filled in the missing bed, bath, house size, and lot size values with the median. I printed and removed 808 duplicate rows where the same listing appeared multiple times. I filtered out implausible values by keeping prices between 1,000 and 50 million dollars and house sizes between 100 and 20,000 square feet. I converted the ZIP code column from a decimal number to a 5 digit string so it would match the Census format. After cleaning I had 2,221,128 listings and 99.3 percent of them matched a Census record when I merged the datasets by ZIP code. I engineered three new features. Price per square foot lets me compare homes of different sizes. Listing count per ZIP measures market activity. Net domestic migration by state measures which markets are gaining or losing people.

Exploratory Data Analysis:

The EDA revealed several patterns that shaped the modeling. Price is heavily right skewed with a skewness of 15.45. The median price is 325,000 dollars while the mean is 515,234 dollars because luxury listings pull the average up. Log transforming price brings the skewness down to

-0.57, so I used log price as the modeling target. The typical listing is a 3 bedroom, 2 bathroom home of about 1,760 square feet on a quarter acre at roughly 180 dollars per square foot. Bedroom count does not raise prices in a straight line. Prices stay flat from 1 to 3 bedrooms and climb sharply starting at 4. Location proved as important as the property itself. Hawaii, California, and the District of Columbia have the highest median prices and price per square foot. ZIP code median household income correlates with price at 0.28. The migration analysis showed Texas, Florida, North Carolina, and Arizona gained people every year from 2019 to 2024 while California, New York, and Illinois lost people every year. California lost over 255,000 net residents in 2024 alone. People are leaving the most expensive states for more affordable Sunbelt markets.

Pre processing:

I prepared the data for machine learning in four steps. I used log price as the target. I created dummy features for the two categorical columns, status and state, which grew the feature set from 11 columns to 62. I standardized the nine numeric features with StandardScaler fit on the training data only, so no information from the test set leaked into training. I split the data 80/20 into 238,372 training rows and 59,593 testing rows. Because the full dataset is 2.2 million rows, I worked with a random sample of 300,000 rows so the models could train in a reasonable time. I dropped price per square foot from the features because it is calculated from price and would leak the answer into the model.

Modeling:

I built three models and tuned the best one. Linear regression served as the baseline. Ridge regression added regularization to handle the correlation I found between bed, bath, and house size. Random forest captured non linear patterns like the bedroom price jump at 4 bedrooms. The random forest clearly outperformed the linear models, so I tuned it with GridSearchCV using 3 fold cross validation over tree count and tree depth. The best settings were 100 trees with a max depth of 20. All models were evaluated on the same held out test set using RMSE and R squared.

Model	RMSE	R squared
Linear Regression	0.8491	0.4341
Ridge Regression	0.8491	0.4341
Random Forest	0.6014	0.7161
Tuned Random Forest (final)	0.5661	0.7485

The tuned random forest is the final model. It explains about 75 percent of the variation in log listing price, a large improvement over the 43 percent explained by the linear models. The actual versus predicted scatter plot shows most predictions clustering tightly around the perfect prediction line.

Findings:

The feature importances from the final model confirm the story from the EDA. Bathrooms are the single most important feature at 0.28, followed by ZIP code median household income at 0.17, house size at 0.16, and lot size at 0.13. State dummies for California and New York also carry meaningful weight. Together this means a home price is driven roughly equally by what the home is and where it sits. Two identical homes can differ enormously in price purely because of the wealth of the surrounding ZIP code and the state market they are in.

Recommendations:

First, buyers and agents should benchmark a listing against its ZIP code income level, not just comparable homes. A listing priced far above what its ZIP income supports is likely overpriced and the model can flag those gaps automatically.

Second, investors seeking value should focus on states with positive net migration and low price per square foot such as Texas, North Carolina and Tennessee. These markets combine growing demand with affordable entry prices while high outflow states like California and New York face demand headwinds despite high prices.

Third, sellers should highlight bathrooms in listings and consider bathroom additions before selling. Bathrooms carry the highest feature importance in the model ahead of overall square footage, so an added bathroom moves predicted price more than most other improvements.

Ideas for Further Research:

The next step is to make the model forward looking. I plan to pull ACS data for 2019 and 2023 at the ZIP level and calculate the change in median household income, educational attainment and demographics between those two years. That trajectory becomes a new set of features showing how fast and in which direction each ZIP code is moving. Combined with the migration trends, the goal is a model that ranks ZIP codes by predicted appreciation over the next five years. Other improvements include adding gradient boosting models like XGBoost, testing the model on the full 2.2 million rows with more computing power and adding neighborhood level features like school ratings and walkability that the current dataset lacks.